CS & IT ENGINEERING

Theory of Computation

DFA Construction

DFA

Lecture No.- 03

By- Venkat sir

Recap of Previous Lecture

Topic — DFA Design

Topic — Possibility of DFA.

if Dependency exist → DFA not possible

no Dependency → DFA possible

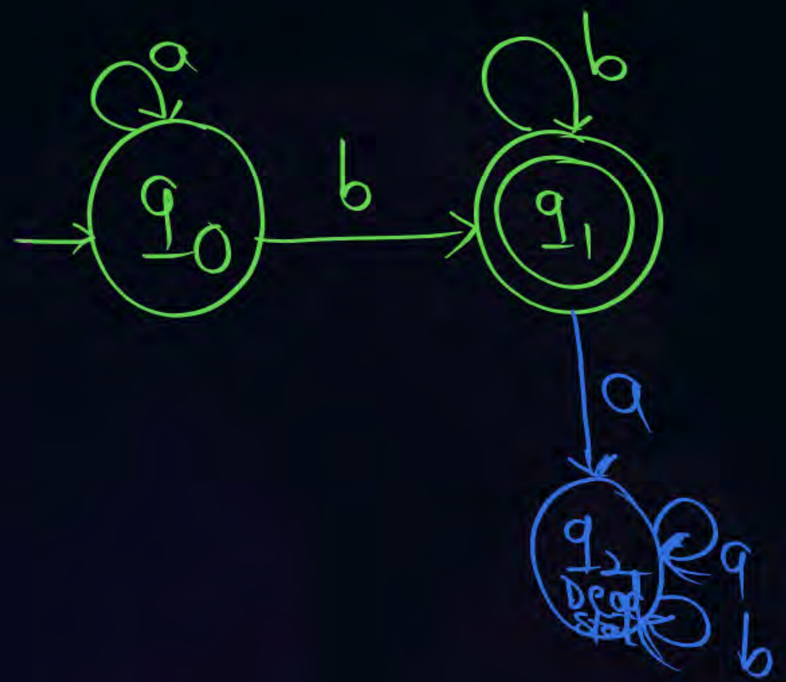
if DFA possible then that language
is known as Regular Language.

if DFA not possible ⇒ Non Regular Language.

FINITE AUTOMATA

(1) Construct DFA for $L = \{a^n b^m \mid n \geq 0, m \geq 1\}$

$L = \{b, b^2, b^3, \ldots$
$ab, abb, aab \ldots\}$



3 States

(Q) Construct DFA for $L = \{a^n b^m / n, m \geq 0\}$ Home Work

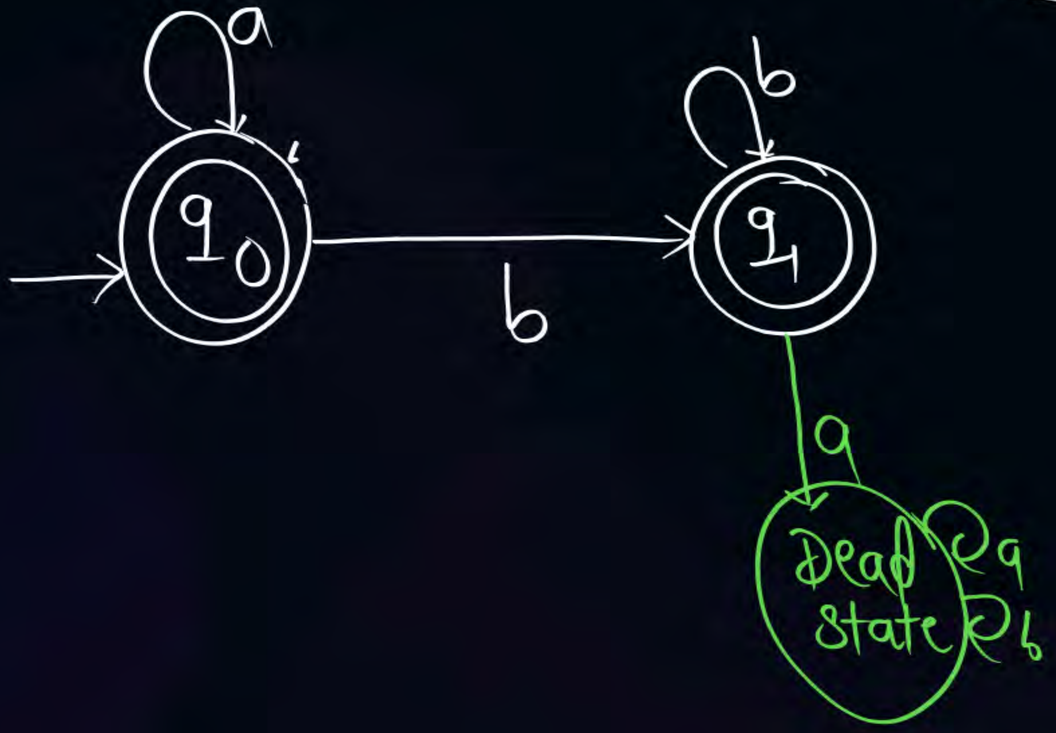
ba ×

ϵ

$a, a^2, a^3 \ldots$

$b, b^2, b^3 \ldots$

$ab, ab^2, a^2b \ldots$



$q_0 = \{\epsilon\}$

$q_1 = \{\}$

(1) Construct DFA for $L = \{a^{2n} \mid n \geq 1\}$ $\Sigma = \{a\}$

$\{a^2, a^4, a^6, a^8, a^{10} \cdots\}$

$\{a^2, a^2 \cdot a^2, a^2 \cdot a^2 \cdot a^2, a^2 \cdot a^2 \cdot a^2 \cdot a^2, \cdots\}$

DFA ?

(1) Construct DFA for $L = \{a^{n^2} / n \geq 1\}$

$\{a^1, a^4, a^9, a^{16} ---\}$

$\{$ DFA Not possible $\}$

ⓐ possible

ⓑ Not possible

NOTE

If a language formed over one symbol then Common difference should exist to Construct DFA.

(Q) Which of the following language is Regular ? {DFA possible}

(a) $L = \{a^{2^n} \mid n \geq 1\} = \{a^2, a^4, a^8, a^{16} \ldots\}$ × Non Regular

$L = \{a^n b^m \mid n > m\}$ — Non Regular

(b) $L = \{a^{n^2} \mid n \geq 1\} = \{a^1, a^4, a^{27} \ldots\}$ × Non Regular

(c) $L = \{a^p \mid p \text{ is prime number}\}$ → Non Regular

(d) None

Topic : Finite Automata

It is a mathematical model which contains finite number of states and transitions.

Topic : Deterministic Finite Automata

DFA : It is a finite automata in which from every state on every input symbol exactly one transition should exits.

Topic : Deterministic Finite Automata

FORMAL DFA : DFA is defined as

$DFA = (Q, \Sigma, q_0, F, \delta)$

Q : Finite set of states

Σ : Input alphabet

q_0 : Initial state

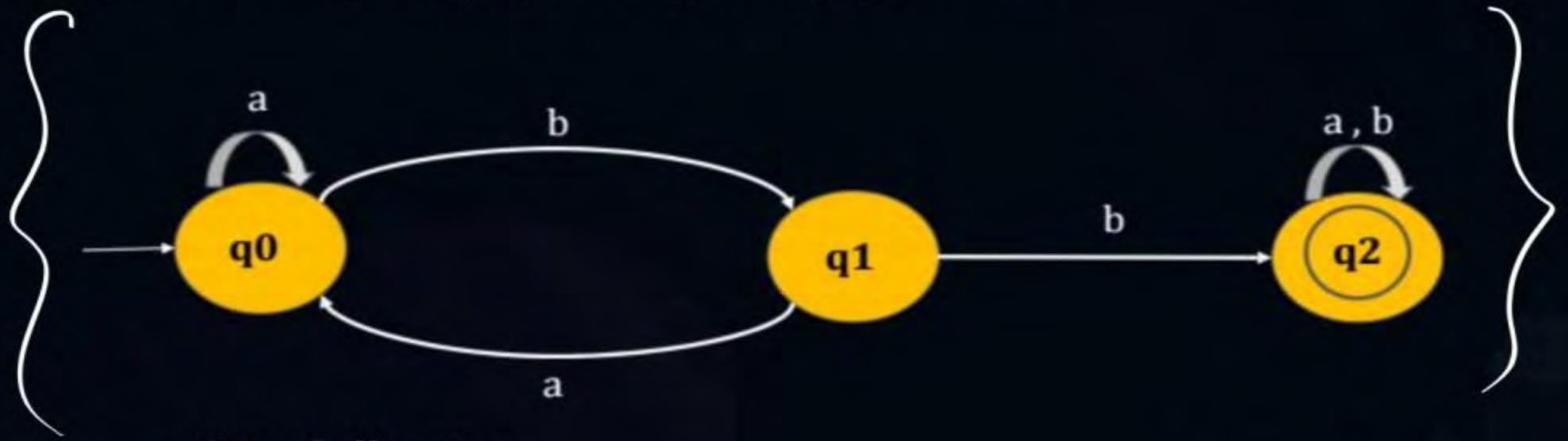
F : Set of final states

δ : Transition function $Q * \Sigma \rightarrow Q$

#Q. Identify language accepted by given DFA

Sub string bb

bab



Set of all strings

A Starting with bb

B Ending with bb

C Contains at least 2 b's

D None

#Q. Identify the language accepted by following DFA

$(0+1)^*00(0+1)^*$

$\Sigma = \{0, 1\}$

ending with 00

elimination

set of all strings

A ✗ Starting with 00 = {00, 000, 0001 - - -}

B ✗ Substring 00 {00, 001, 100, 1001 - - -}

C ✗ Ending with 0 = {0, 00, 10 - - -}

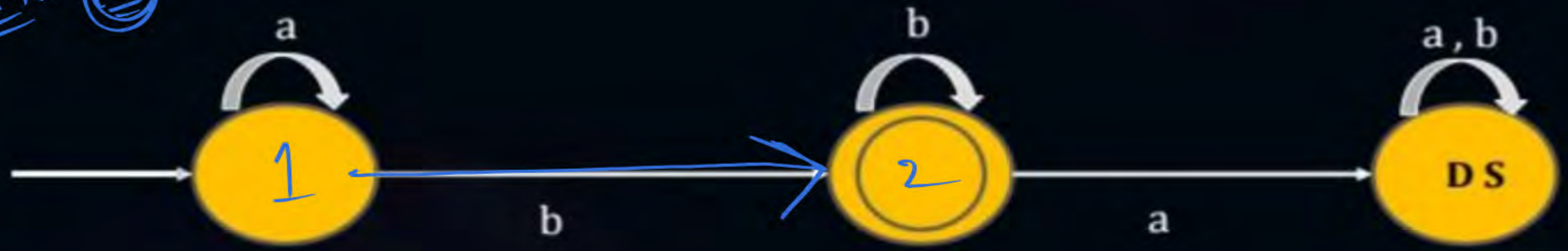
D ✓ None

{b, bb, ab ---}

elimination

#Q. Identify language accepted by following DFA

DFA = {b ---}



A ✗ {ab} $L = \{a^n b^m \mid n, m \geq 1\}$

B ✗ {@ ---} $L = \{a^n b^m \mid n \geq 1, m \geq 0\}$

C ✗ $L = \{a^n b^m \mid n, m \geq 0\} = \{\epsilon,$

D None

#Q. Identify language accepted by following DFA

$L = \{\epsilon, _____\}$ Home Work

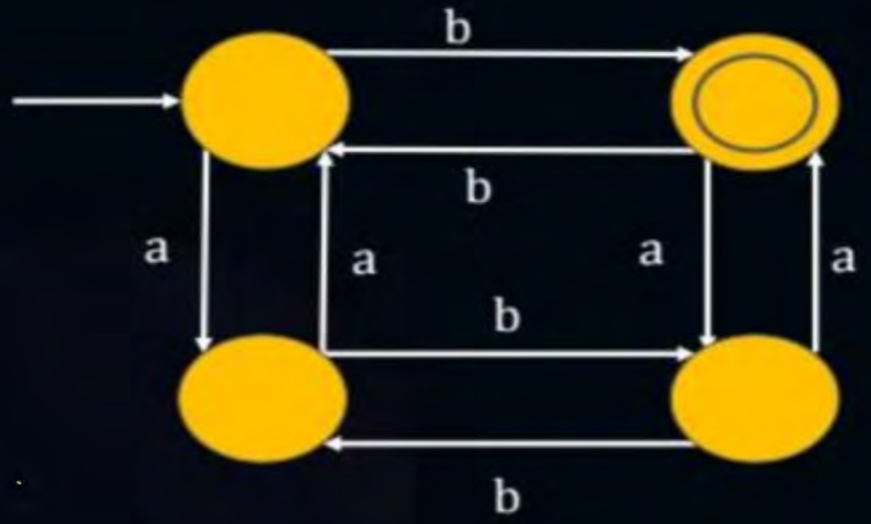
A $L = \{a^n b^m \mid n, m \geq 1\}$

B $L = \{a^n b^m \mid n \geq 1, m \geq 0\}$

C $L = \{a^n b^m \mid n, m \geq 0\}$

D None

#Q. Identify language accepted by following DFA



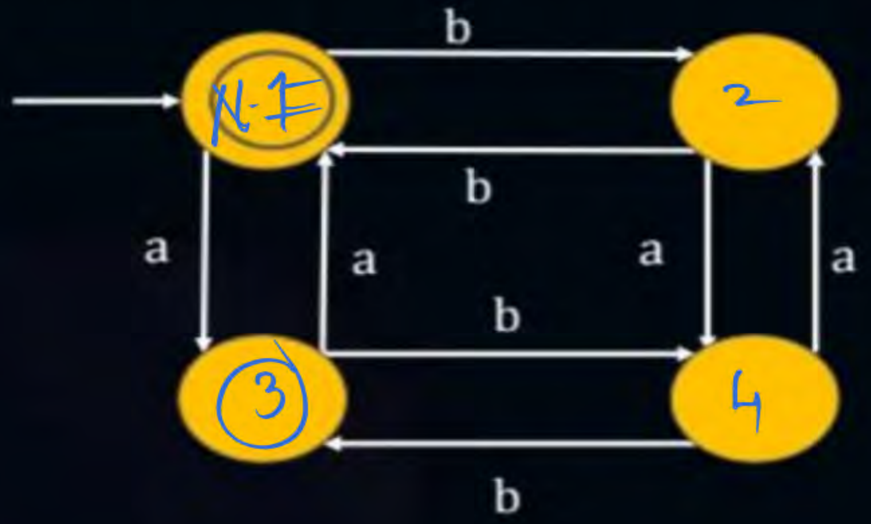
A # a's even and # b's even

B # a's odd and # b's even

C # a's odd and # b's odd

D # a's even and # b's odd

#Q. Identify language accepted by following DFA



elimination

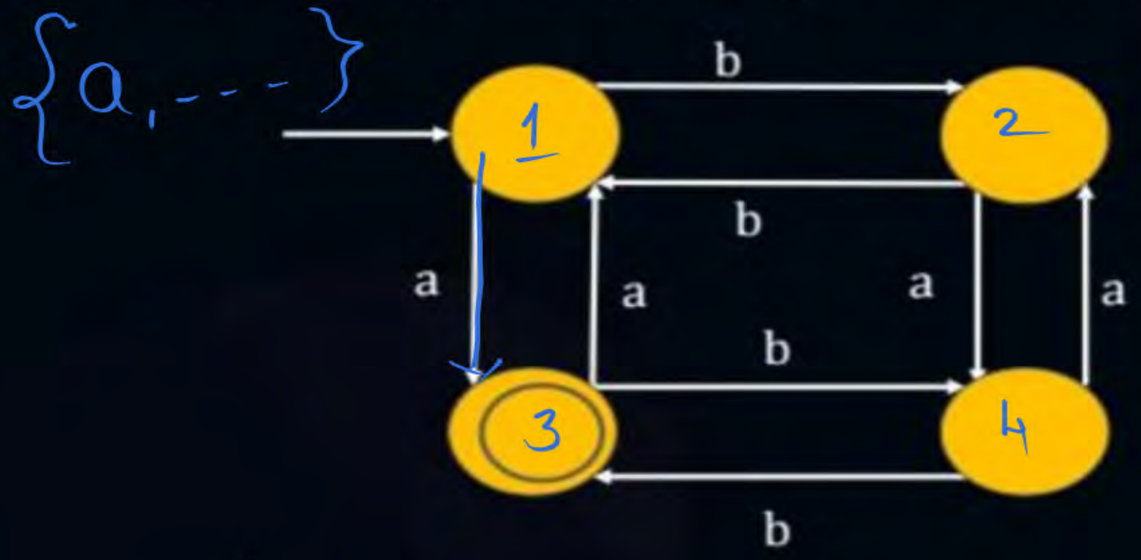
A # a's even and # b's even

B # a's odd and # b's even

C # a's odd and # b's odd

D # a's even and # b's odd

#Q. Identify language accepted by following DFA



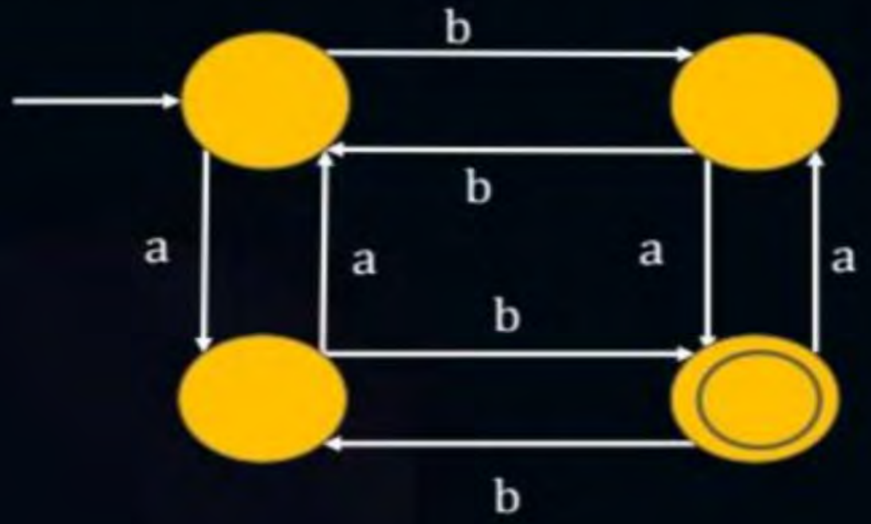
A # a's even and # b's even

B # a's odd and # b's even

C # a's odd and # b's odd

D # a's even and # b's odd

#Q. Identify language accepted by following DFA



A # a's even and # b's even

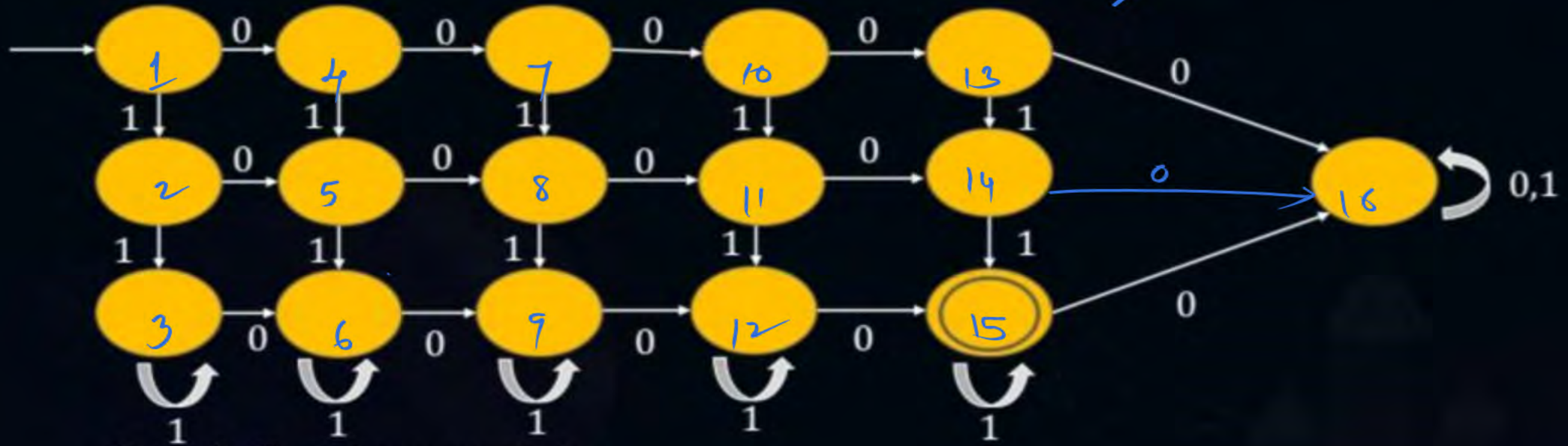
B # a's odd and # b's even

C # a's odd and # b's odd

D # a's even and # b's odd

#Q. Identify language accepted by following DFA

Home Work



Set of all strings contains

A Length of the string alteast 6

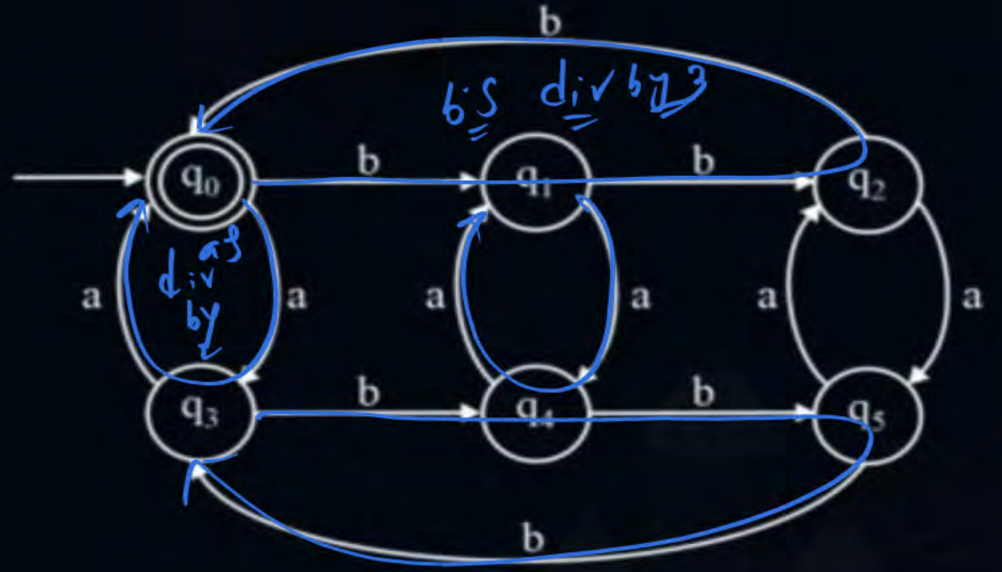
B # 0's exactly4 and 1's atleast 3

C # 0's atleast 4 and # 1's exactly 2

D None

MCQ

#Q. Consider the finite automaton m, m accepts all strings of a's and b's m which the number of a's and b's are respectively.



A Divisible by 3 and 2

B Odd and even

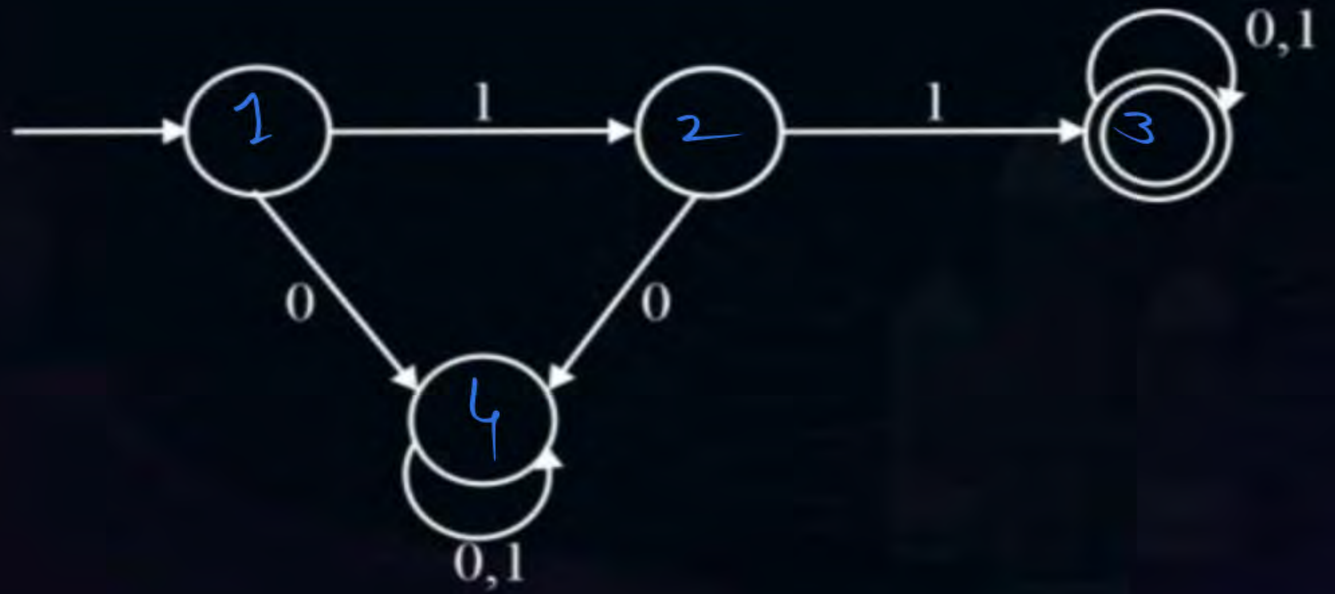
C Even and odd

D Divisible by 2 and 3

MCQ

Home work

#Q. The number of binary strings of length n' accepted by the following finite automata is-



A $\{2^n | n \geq 0\}$

B $\{2^{n-1} | n \geq 1\}$

C $\{2^{n-2} | n \geq 2\}$

D $\{2^{n-3} | n \geq 3\}$

#Q. For which of the following languages DFA is possible

1. $L = \{a^n b^m \mid n,\ m \geq 1\}$

2. $L = \{a^n b^n \mid n \geq 1\}$

3. $L = \{a^n b^m \mid n < m\}$

4. $L = \{a^n b^m \mid\ n \neq m\}$

5. $L = \{a^n b^m c^{n+m} \mid n,\ m \geq 1\}$

6. $L = \{a^n b^{2m} \mid n,\ m \geq 1\}$

MCQ

#Q. Which of the following are regular sets?

1. $\{a^n b^{2m} \mid n \geq 0, m \geq 0\}$
2. $\{a^n b^m \mid n = 2m\}$
3. $\{a^n b^m \mid n \neq m\}$
4. $\{x\, c\, y \mid x, y \in \{a, b\}^*\}$

A 1 and 4 only

B 1 and 3 only

C 1 only

D 4 only

Topic : DFA Construction

Regular Language

Construct DFA for the following Language.

{ set }

1. $L = \{a^n b^m \mid n, m \geq 1\}$ → Regular Language ✓
2. $L = \{a^n b^n \mid n \geq 1\}$
3. $L = \{a^n b^m \mid n < m\}$
4. $L = \{a^n b^m \mid n \neq m\}$ ⟶ non Regular
5. $L = \{a^n b^m c^{n+m} \mid n, m \geq 1\}$
6. $L = \{a^n b^{2m} \mid n, m \geq 1\}$ ⟶ Regular. ✓

Topic : DFA Construction

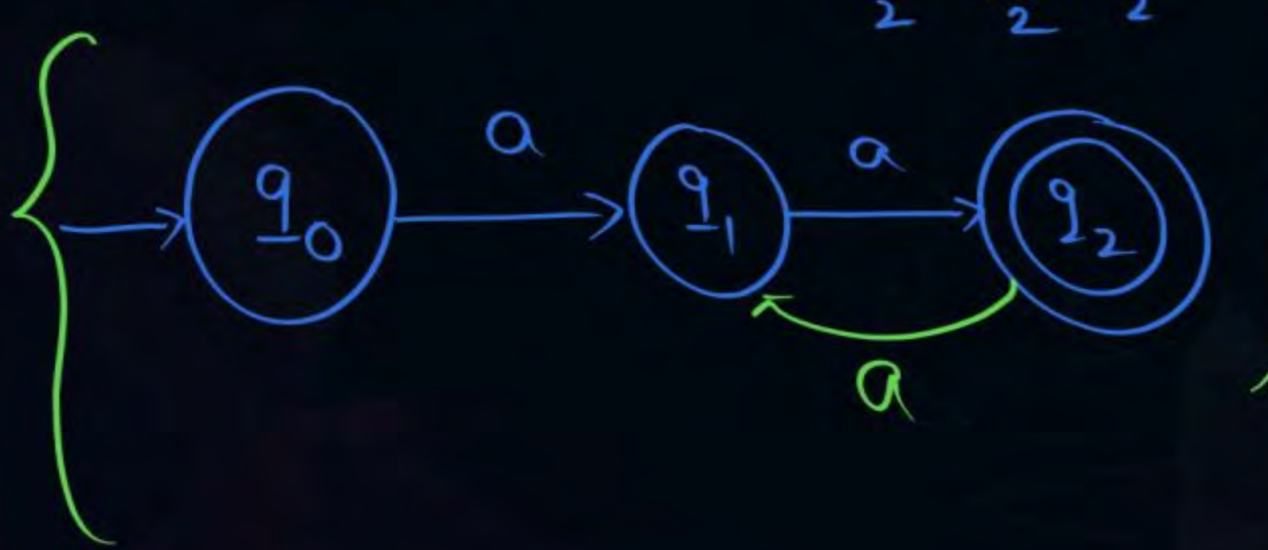
$L = \{a^{2n} \mid n \geq 1\}$ $\Sigma = \{a\}$

Construct DFA for the following Language.

$\{a^2, a^4, a^6, a^8 \text{----}\}$

2 2 2

3 states



Topic : DFA Construction

Construct DFA for the following Language.

$L = \{ a^{3n} \mid n \geq 1 \}$

$\{ a^3, a^6, a^9, a^{12} \ldots \}$

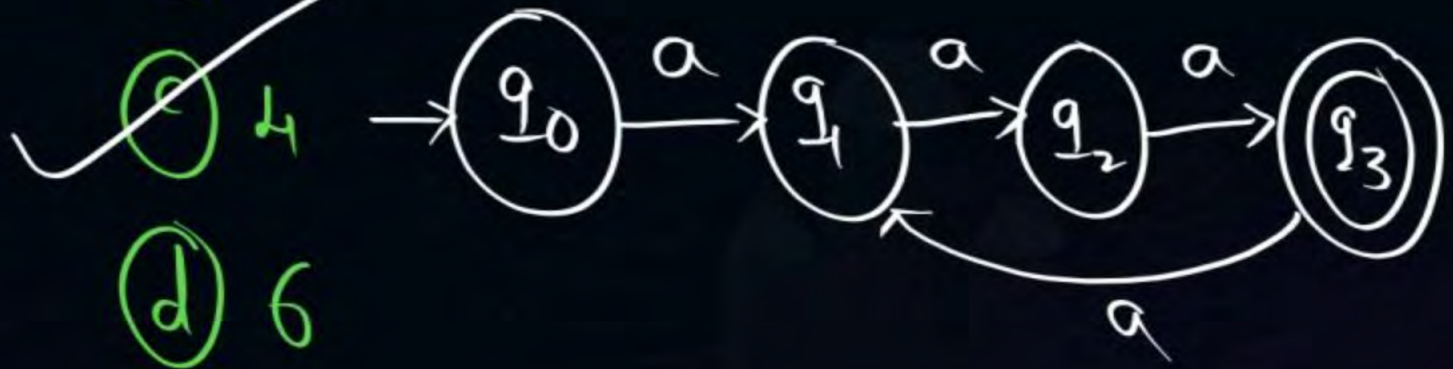
3 3 3

(a) 2

(b) 3

(c) 4

(d) 6



$L = \{a^{2^n} \mid n \geq 1\}$ ↩ DFA not possible

How many states in DFA

(a) 2 (b) 4 (c) 8 (d) none

$\{a^2, a^4, a^8, a^{16} \ldots\}$

Telegram
@ venkatsirpw

THANK - YOU